James Oswald Nde Naoussi

Google DeepMind Scholar | Physics & AI for Science

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james-naoussi | dev-svald | French & English (bilingual)

PROFESSIONAL PROFILE

Physicist and former Google DeepMind Scholar from the AI for Science stream at the African Institute for Mathematical Sciences (AIMS), South Africa. I combine a rigorous training in mathematics and physics with a growing specialisation in machine learning, artificial intelligence, and quantum computing. I am motivated by teaching, clear scientific communication, and building solutions that translate theory into measurable impact. My goal is to strengthen my expertise across mathematics, science, and the use of AI to tackle complex, real-world problems with creativity and rigour. Having completed my Master's in Mathematical Sciences at AIMS South Africa, I am open to PhD positions in the fields of Machine Learning optimization, Human–AI collaboration, and Applied AI in education, and open to collaborate on AI for Good projects.

EDUCATION

African Institute for Mathematical Sciences (AIMS) — Cape Town, South Africa
AI for Science — MSc in Mathematical Sciences — *Sept. 2025 – July 2026*

- **Recipient** of the Google DeepMind Scholarship.
- **Focus:** bridging the intersection of mathematics, machine learning and AI to address challenges in different science disciplines.
- **Courses included:** Machine learning, Quantum Computing, NLP/LLM, Reinforcement Learning, Matrix and Tensor factorisation, Fluid Dynamics.

University of Buea — Buea, Cameroon
MSc in Physics — *Oct. 2022 – Dec. 2024*

- **Courses included:** Quantum mechanics, Mathematical Methods, Electrodynamics, Continuum Dynamics.
- **GPA:** 3.65

University of Buea — Buea, Cameroon
BSc in Applied Physics — *Oct. 2016 – Dec. 2019*

- **Courses included:** Digital electronics, Communication systems, Microprocessors, Computer Architectures and Data Network.
- **GPA:** 3.20

Bilingual Grammar School Molyko — Buea, Cameroon
Baccalauréat en Sciences Mathématiques — *Sept. 2015 – June 2016*

PROFESSIONAL EXPERIENCE

African Institute for Mathematical Sciences (AIMS) — Cape Town, South Africa
Teaching Assistant — AI for Science Stream — *Sept. 2026 – Present*

Private Tutor — Mathematics & Applied Sciences — Buea, Cameroon
Self-employed; long-term client relationships across secondary and undergraduate levels — *Sept. 2020 – June 2025*

Bilingual Grammar School Molyko — Buea, Cameroon
Secondary School Mathematics Teacher — *Oct. 2022 – June 2023*

RESEARCH PROJECTS

Deep Learning for Quantum Error Mitigation using Classical Shadows — *June 2026 (unpublished)*

Automated Essay Scoring (AES): Human–AI Collaboration with Budget and Fairness Constraints — *Mar. 2026 (unpublished)*

Multilingual and Multicultural Detection of Online Polarization — *Dec. 2025 (unpublished)*

Movie Recommender System using ALS and Matrix Factorization — *Nov. 2025 (unpublished)*

PUBLICATION

High intensity and frequency modulations based on infinite pulse trains and ultrashort pulses — *2025*
Chaos **35**(8)

WORKSHOPS & SHORT PROGRAMMES

Deep Learning Indaba — Lagos, Nigeria
Skills in networking and presenting scientific work — *Aug. 2026*

Africa Scientifique — AIMS South Africa
Skills in scientific communication — *25–27 Feb. 2026*

Theory of Quantum Machine Learning algorithms — AIMS South Africa
Bridging the intersection of Quantum Computing and Machine Learning — *27–30 Oct. 2025*

Creative Genius Hut — Buea, Cameroon
Social & entrepreneurial skills — *23 June – 25 July 2025*

LEADERSHIP, SERVICE & COMMUNITY

AIMS South Africa — Cape Town, South Africa
Head Teaching Assistant — AI for Science Stream — *Sept. 2026 – Present*

AIMS South Africa — Cape Town, South Africa
Class Representative and Club President — *Dec. 2025 – July 2026*

Creative Genius Hut — Buea, Cameroon
Team Lead — *June – July 2025*

Former Parish (Youth Ministry) — Buea, Cameroon
Youth Leader — *Sept. 2020 – Aug. 2025*

CORE COMPETENCIES

- **Scientific computing & ML:** Python, MATLAB, introductory machine-learning workflows, reproducible analysis.
- **Physics & mathematics:** Applied & mathematical physics training; modelling, simulation, and analytical reasoning.
- **Communication:** Classroom instruction, technical writing, science storytelling for non-specialists.
- **Digital collaboration:** Social media for outreach, remote coordination, community building.

BEYOND THE LAB

Personal interests: travelling, hiking, choral singing, reflective reading.
I recharge through nature walks and cultural exploration; music keeps me grounded creatively, while wide reading feeds curiosity across science communication, leadership, and futures thinking.

REFERENCES

Prof. Ulrich Paquet, Director of the AI for Science Master
African Institute for Mathematical Sciences (AIMS), South Africa
ulrich@aims.ac.za

Prof. Dikande Alain Moise, Head of Physics Department
University of Buea, Cameroon
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Prof. Eric Fokam, Head of Zoology and Animal Physiology Department
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